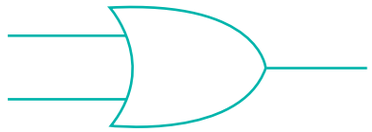
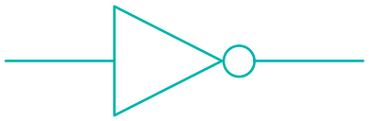
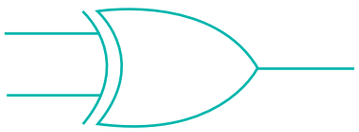
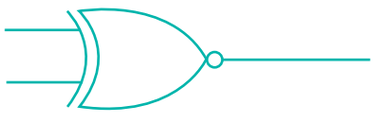
Table 2. 15
Logic Gates and their corresponding Truth Tables

Logical Gates	Symbol	Truth Table
AND		A | B | AB 0 | 0 | 0 0 | 1 | 0 1 | 0 | 0 1 | 1 | 1
OR		A | B | A + B 0 | 0 | 0 0 | 1 | 1 1 | 0 | 1 1 | 1 | 1
NOT		A | $\overline{A}$ 0 | 1 1 | 0
NAND		A | B | $\overline{A\,B}$ 0 | 0 | 1 0 | 1 | 1 1 | 0 | 1 1 | 1 | 0
NOR		A | B | $\overline{A + B}$ 0 | 0 | 1 0 | 1 | 0 1 | 0 | 0 1 | 1 | 0
XOR		A | B | A⊕B 0 | 0 | 0 0 | 1 | 1 1 | 0 | 1 1 | 1 | 0
XNOR		A | B | $\overline{A \oplus B}$ 0 | 0 | 1 0 | 1 | 0 1 | 0 | 0 1 | 1 | 1

11th Std Computer Science_EM Chapter 2.indd 39 06-01-2023 15:02:40